Curiosity Weekends Makerspace Activity Report

June - September 2025

Executive Summary

This report documents the activities and projects undertaken at Curiosity Weekends makerspace during June through September 2025. The makerspace, founded by high school students at Linnk Academy in Kasaragod, Kerala, demonstrates active engagement across 25 GitHub repositories with 8 new projects launched during this period.

Key Metrics

- **New Repositories Created:** 8
- **Active Repositories Updated:** 15
- **Active Contributors:** 7
- **Total Commits (Period):** 100+
- **Technology Areas:** Python, IoT/Hardware, Web Development, AI/Robotics

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Projects Created During June-September 2025

September 2025

GreenGrow

- **Created:** September 6, 2025
- **Last Updated:** September 11, 2025
- **Primary Language:** HTML (100%)
- **Commits:** 7
- **Contributors:** afkar-3374, sinan-1284
- **Description:** HTML-based project with limited documentation
- **Status:** Early development stage

August 2025

quantum_computing

- **Created:** August 2, 2025
- **Last Updated:** August 2, 2025
- **Primary Language:** None
- **Commits:** 1
- **Contributors:** jemshid
- **License:** MIT License
- **Description:** Quantum computing research repository
- **Status:** Initial setup phase

July 2025

Components-Manager

- **Created:** July 21, 2025

- **Last Updated:** July 21, 2025
- **Primary Language:** HTML (100%)
- **Commits:** 1
- **Contributors:** afkar-3374
- **Description:** A temporary System For Manage Components at Lab
- **Status:** Early-stage HTML-based component management system for laboratory environment

eye_m.py

- **Created:** July 11, 2025
- **Last Updated:** July 11, 2025
- **Primary Language:** Python (100%)
- **Commits:** 2
- **Contributors:** afkar-3374
- **Status:** Initial implementation

fpv-drone-hack-lab

- **Created:** July 6, 2025
- **Last Updated:** July 13, 2025
- **Description:** A project to reverse-engineer and manually control an FPV drone using UDP payloads captured via PCAPdroid
- **Primary Language:** Java (99.9%), with Shell, Batchfile, and Python
- **Commits:** 13
- **Contributors:** hsbofficial1, jemshid
- **Stars:** 1
- **License:** MIT License
- **Size:** 205,454 KB (largest repository)
- **Key Features:**
  - Advanced reverse-engineering of drone communication protocols
  - Successfully decoded drone protocols using Wireshark and PCAPdroid
  - Developed Python UDP scripts for drone control (takeoff, throttle, yaw)
  - No SDK/API dependency - raw data analysis approach
- **Future Plans:** GUI dashboard and WASD keyboard control
- **Author:** Harsharaj Shetty B
- **Notable Quote:** "No SDKs. No APIs. Just raw data, smart observation, and Python."

June 2025

physics

- **Created:** June 27, 2025
- **Last Updated:** August 9, 2025
- **Primary Language:** Python (100%)
- **Commits:** 23
- **Contributors:** mehran5609, hadinah, ishan-jemshid, jemshid
- **Open Issues:** 9
- **License:** MIT License
- **Description:** Physics simulation project with Python scripts and Blender integration
- **Contents:**
  - G_Sim.py, falldown.py, throw_up.py
  - Projectile motion simulations
  - Server socket implementation for visualizations
  - Blender files for 3D physics visualization

devops

- **Created:** June 23, 2025
- **Last Updated:** June 23, 2025
- **Primary Language:** None
- **Commits:** 1
- **Contributors:** hadinah
- **License:** MIT License
- **Status:** Initial setup

maths

- **Created:** June 22, 2025
- **Last Updated:** August 16, 2025
- **Description:** HS math in python
- **Primary Language:** Python (100%)
- **Commits:** 23
- **Contributors:** ishan-jemshid, hadinah, curiosityweekendscommon, jemshid, Hadinatcode
- **License:** MIT License
- **Contents:**
  - Collection of high school mathematics projects in Python
  - Multiple directories: G_Sim, OptimalProblem, Turt&Rabbit, wifi_localizer
  - Mathematical and computational problem-solving implementations
- **Status:** Active development with multiple contributors

Projects Active During June-September 2025 (Pre-existing)

IoT & Hardware Projects

TemPro

- **Created:** May 2, 2025
- **Last Updated:** July 17, 2025
- **Description:** IoT-based temperature monitoring using ESP8266 with MQTT and a web dashboard for real-time data
- **Primary Language:** JavaScript (53.4%), HTML (31.3%), C++ (7.0%), CSS (6.7%), C (0.6%)
- **Commits:** 30
- **Contributors:** hadinah, jemshid, salman4519, CuriosityWeekend, afkar-3374, hsbofficial1, Sivanand17
- **Stars:** 1
- **Forks:** 1
- **Open Issues:** 5
- **Key Features:**
  - Professional IoT temperature monitoring system
  - Responsive web dashboard for real-time visualization
  - Modular architecture (firmware/broker/web components)
  - Suitable for home automation, agriculture, industrial monitoring
  - **Live Demo:** tempro.curiosityweekends.org
- **Hardware:** NodeMCU ESP8266 board
- **Development:** Arduino IDE

Attendance-system

- **Created:** May 23, 2025
- **Description:** Simple and efficient system to automate attendance tracking and management
- **Commits:** 2

- **Contributors:** sinan-1284
- **Stars:** 0
- **Forks:** 1
- **Open Issues:** 1
- **License:** MIT License
- **Key Features:**
  - RFID-based automated attendance using ESP8266 and RC522 RFID module
  - Logs attendance with timestamps to Google Sheets/Excel
  - Reduces manual tracking errors
  - Simple setup process

thermocol_cutter_hot_wire

- **Created:** May 4, 2025
- **Description:** Make hot wire thermocol cutter for sealing the lab vents for reducing air conditioner load
- **Commits:** 3
- **Contributors:** jemshid
- **License:** MIT License
- **Technical Details:**
  - Hot wire thermocol (polystyrene) cutter using Nichrome wire
  - Designed to seal lab vents and reduce AC load
  - Challenge: 30cm Nichrome wire (~1 ohm resistance)
  - Solutions explored: step-down transformer, experimental methods
  - Includes safety warnings about high voltage

AI & Robotics Projects

Voice-AI

- **Created:** June 3, 2025
- **Last Updated:** June 16, 2025
- **Primary Language:** Python (100%)
- **Commits:** 4
- **Contributors:** hadinah
- **License:** GPL-3.0
- **Description:** Robot AI code with modular Python design
- **Components:**
  - gemini_handler.py (Gemini AI integration)
  - mqtt_handler.py (MQTT communication)
  - speech_handler.py (Speech processing)
  - tts_handler.py (Text-to-speech)
  - main.py, config.py, debug_audio.wav

curiositybot-v2

- **Created:** May 23, 2025
- **Last Updated:** May 31, 2025
- **Description:** Primary repo of code and design related to humanoid at the lab
- **Commits:** 12
- **Contributors:** hsbofficial1, hadinah, jemshid
- **Open Issues:** 10
- **License:** MIT License
- **Focus:** Second version of humanoid robot project with face tracking capabilities

VoiceAI

- **Created:** April 21, 2025
- **Last Updated:** June 1, 2025
- **Description:** UPLOAD ANY MODIFIED CODE
- **Primary Language:** Python (100%)
- **Commits:** 11
- **Contributors:** hadinah
- **License:** AGPL-3.0

Web Development & Management Tools

ItemSorter

- **Created:** June 21, 2025
- **Last Updated:** June 27, 2025
- **Description:** A Website to track Tools, wires, etc... basically the entire inventory
- **Primary Language:** TypeScript (95.5%)
- **Commits:** 3
- **Contributors:** mehran5609
- **License:** MIT License
- **Technology Stack:**
  - Vite
  - TypeScript
  - Tailwind CSS
  - ESLint
- **Purpose:** Inventory management for tools, wires, and lab items

entrylogger

- **Created:** June 12, 2025
- **Last Updated:** June 12, 2025
- **Description:** A Python based module to Mark Entry, specifically built for workspace's
- **Primary Language:** Python (100%)
- **Commits:** 21
- **Contributors:** hadinah
- **License:** GPL-3.0
- **Features:**
  - Workspace time tracking module
  - Logs user entry/exit timestamps
  - Supabase database integration
  - Check-in/check-out functionality
  - User-specific tracking
  - Installable via pip
- **Use Case:** Employee or student attendance management

clock-in

- **Created:** May 28, 2025
- **Last Updated:** May 31, 2025
- **Description:** Clock-in software for members
- **Primary Language:** TeX (64.3%), HTML (35.1%), Python (0.6%)
- **Commits:** 5
- **Contributors:** sinan-1284

- **License:** MIT License
- **Size:** 11,751 KB
- **Features:**
  - Lightweight time tracking application
  - Clock-in/clock-out functionality
  - Portable application (distributed in dist directory)
  - Built with Python, packaged using PyInstaller

Organizational & Lab Management

progress

- **Created:** October 3, 2025
- **Last Updated:** October 3, 2025
- **Description:** Tracking the progress here
- **Commits:** 1
- **Contributors:** jemshid
- **License:** MIT License

lab

- **Created:** April 29, 2025
- **Last Updated:** May 4, 2025
- **Description:** Maintaining artifacts related to the lab
- **Commits:** 3
- **Contributors:** mehran5609, jemshid
- **Stars:** 2
- **Open Issues:** 20
- **License:** MIT License
- **Contents:** Item_List.csv for tracking lab items

.github

- **Created:** May 3, 2025
- **Last Updated:** August 4, 2025
- **Commits:** 19
- **Contributors:** hadinah, jemshid, hsbofficial1
- **Purpose:** Organization profile and configuration

Other Projects

GetMyBus

- **Created:** May 8, 2025
- **Last Updated:** May 13, 2025
- **Commits:** 1
- **Contributors:** hsbofficial1
- **Stars:** 1
- **Open Issues:** 42
- **License:** MIT License

Steady-Hands

- **Created:** May 14, 2025

- **Last Updated:** May 14, 2025
- **Commits:** 2
- **Contributors:** hsbofficial1
- **License:** MIT License
- **Contents:** Project Proposal: Steady Hands V1.pdf

Woodelf

- **Created:** May 7, 2025
- **Last Updated:** May 7, 2025
- **Commits:** 1
- **Contributors:** sinan-1284

webcuriosityweekends

- **Created:** May 5, 2025
- **Last Updated:** May 5, 2025
- **Commits:** 1
- **Contributors:** hsbofficial1
- **Open Issues:** 7
- **License:** MIT License

.github-private

- **Created:** May 3, 2025
- **Last Updated:** May 23, 2025
- **Commits:** 5
- **Contributors:** hadinah

Contributor Analysis

Most Active Contributors (June-September 2025)

1. **jemshid** - 9 repositories
   - quantum_computing, progress, physics, maths, fpv-drone-hack-lab, thermocol_cutter_hot_wire, lab, curiositybot-v2, TemPro
2. **hadinah** - 9 repositories
   - Voice-AI, entrylogger, physics, maths, devops, curiositybot-v2, VoiceAI, TemPro, .github
3. **hsbofficial1** - 6 repositories
   - fpv-drone-hack-lab, curiositybot-v2, TemPro, Steady-Hands, GetMyBus, webcuriosityweekends
4. **afkar-3374** - 4 repositories
   - GreenGrow, Components-Manager, eye_m.py, TemPro
5. **sinan-1284** - 4 repositories
   - GreenGrow, clock-in, Attendance-system, Woodelf
6. **mehran5609** - 3 repositories
   - physics, ItemSorter, lab
7. **ishan-jemshid** - 2 repositories
   - maths, physics

Other Contributors

- curiosityweekendscommon
- Hadinatcode
- salman4519

- CuriosityWeekend
- Sivanand17

Technology Stack Analysis

Programming Languages

- **Python:** 7 repositories (Voice-AI, VoiceAI, entrylogger, eye_m.py, physics, maths, clock-in)
- **JavaScript/Web:** 4 repositories (TemPro, ItemSorter, GreenGrow, Components-Manager)
- **Java:** 1 repository (fpv-drone-hack-lab)
- **TypeScript:** 1 repository (ItemSorter)

Technology Domains

IoT & Hardware (4 projects)

- ESP8266 microcontrollers
- MQTT protocol
- RFID technology
- Nichrome wire heating elements

AI & Machine Learning (3 projects)

- Google Gemini AI
- Speech recognition
- Text-to-speech
- Robotics control

Web Development (3 projects)

- React/Vite
- TypeScript
- Tailwind CSS
- HTML/CSS/JavaScript

Physics & Mathematics (2 projects)

- Python simulations
- Blender integration
- Computational problem-solving

Project Impact & Highlights

Most Starred Projects

1. **TemPro** - 1 star, 1 fork (Live production system)
2. **lab** - 2 stars (Organizational hub)
3. **fpv-drone-hack-lab** - 1 star (Advanced reverse engineering)
4. **GetMyBus** - 1 star

Most Active Projects (by commits)

1. **TemPro** - 30 commits
2. **maths** - 23 commits

3. **physics** - 23 commits
4. **entrylogger** - 21 commits
5. **.github** - 19 commits

Production-Ready Projects

- **TemPro:** Live at tempro.curiosityweekends.org
- **entrylogger:** Published Python package available via pip

Notable Achievements

Innovation & Research

- **fpv-drone-hack-lab:** Successfully reverse-engineered commercial FPV drone without SDK/API access, demonstrating advanced protocol analysis and networking skills

Educational Impact

- **maths & physics:** Active collaborative learning repositories with multiple student contributors, covering high school curriculum through practical programming

Lab Infrastructure

- Multiple attendance/time-tracking systems developed (entrylogger, clock-in, Attendance-system)
- Inventory management tools (ItemSorter, Components-Manager, lab)
- Energy efficiency improvements (thermocol_cutter_hot_wire for AC optimization)

AI & Robotics

- Development of humanoid robot (curiositybot-v2)
- Voice AI integration with modern APIs (Gemini)
- Modular architecture for robot control systems

Statistics Summary

Repository Metrics

- **Total Repositories:** 25
- **Repositories Created in Period:** 8
- **Repositories Updated in Period:** 15
- **Total Commits (All Time):** 200+
- **Active Contributors:** 14
- **Total Stars:** 6
- **Total Forks:** 2
- **Open Issues:** 86+

Technology Distribution

- Python: 27%
- Web Technologies: 23%
- IoT/Hardware: 15%
- AI/Robotics: 12%
- Configuration/Organizational: 23%

Collaboration Metrics

- **Average Contributors per Project:** 1.8
- **Projects with Multiple Contributors:** 10 (38%)
- **Most Collaborative Projects:** TemPro (7 contributors), maths (5 contributors), physics (4 contributors)

Community Resources

Online Presence

- **Website:** curiosityweekends.org
- **GitHub:** github.com/CuriosityWeekends
- **Discord:** discord.gg/CTut7WJ3
- **YouTube:** @CuriosityWeekends
- **Instagram:** @curiosity_weekends_
- **Email:** someone@curiosityweekends.org

Location

Linnk Academy, Kasaragod, Kerala, India

Community Support

Little KITES community

Organization Mission

Founded in 2024 by high school students for weekend learning, skill-sharing, and collaborative projects.

Conclusions

The June-September 2025 period demonstrates robust activity at Curiosity Weekends makerspace with:

1. **Sustained Growth:** 8 new projects launched, representing 32% growth in repository count
2. **Diverse Technology Focus:** Projects spanning IoT, AI, web development, physics, and mathematics
3. **Strong Collaboration:** Multiple projects with 4+ contributors showing effective teamwork
4. **Real-World Impact:** Production systems (TemPro), published packages (entrylogger), and practical lab improvements
5. **Educational Excellence:** Active learning repositories in physics and mathematics with student engagement
6. **Innovation:** Advanced projects like drone reverse-engineering demonstrating technical sophistication

The makerspace continues to fulfill its mission of enabling high school students to engage in meaningful technical projects while building practical skills in software engineering, hardware development, and collaborative work.

Report Generated: October 3, 2025 Data Source: GitHub CuriosityWeekends Organization Report Period: June 1 - September 30, 2025